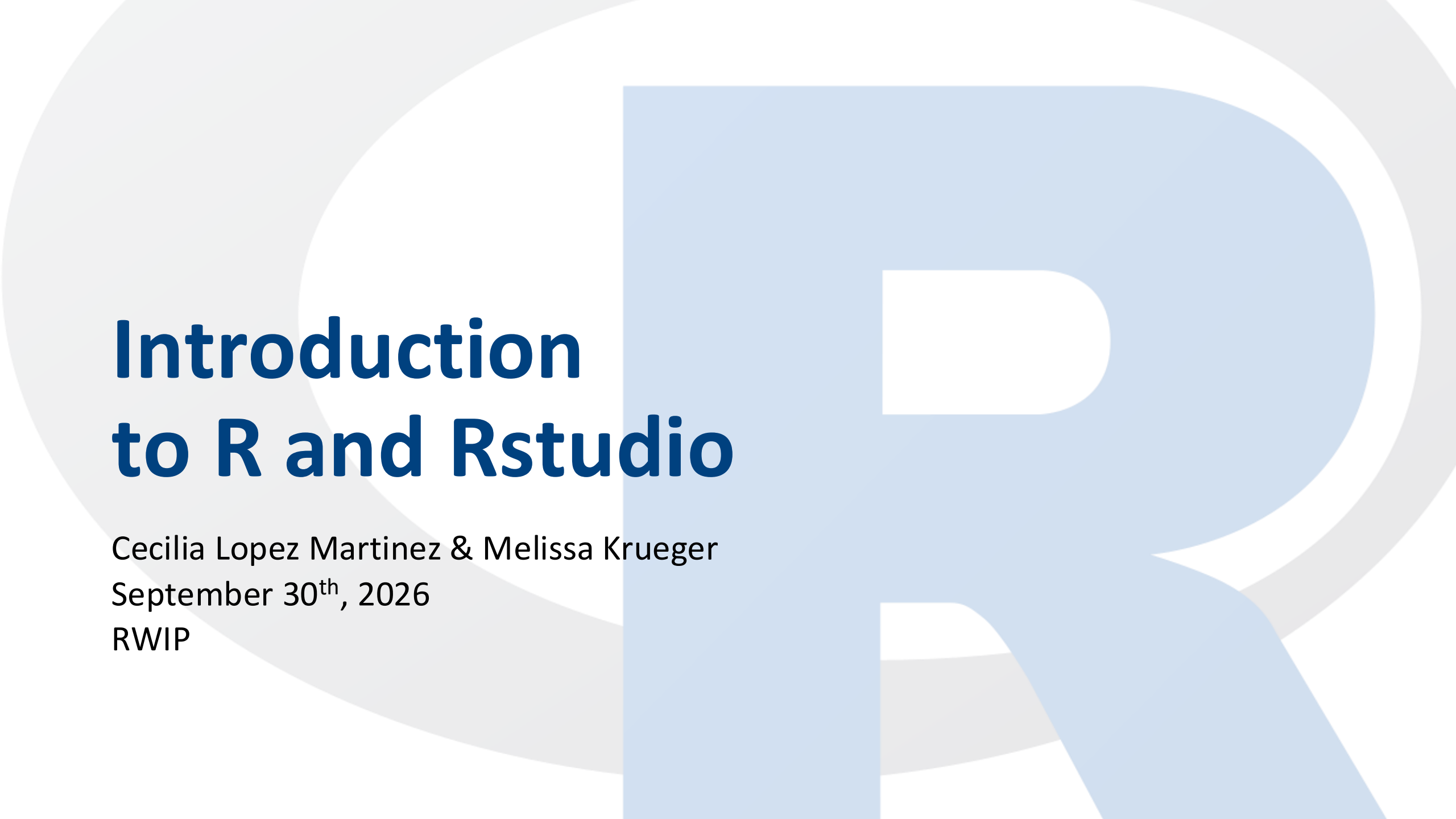




Introduction to R and Rstudio

Cecilia Lopez Martinez & Melissa Krueger

September 30th, 2026

RWIP

What is R?

A computational language for manipulating, statistically analyzing and visualizing data

DATASET

	mpg	cyl	disp
Mazda RX4	21.0	6	160.0
Mazda RX4 Wag	21.0	6	160.0
Datsun 710	22.8	4	108.0
Hornet 4 Drive	21.4	6	258.0
Hornet Sportabout	18.7	8	360.0
Valiant	18.1	6	225.0
Duster 360	14.3	8	360.0
Merc 240D	24.4	4	146.7
Merc 230	22.8	4	140.8
Merc 280	19.2	6	167.6
Merc 280C	17.8	6	167.6
Merc 450SE	16.4	8	275.8

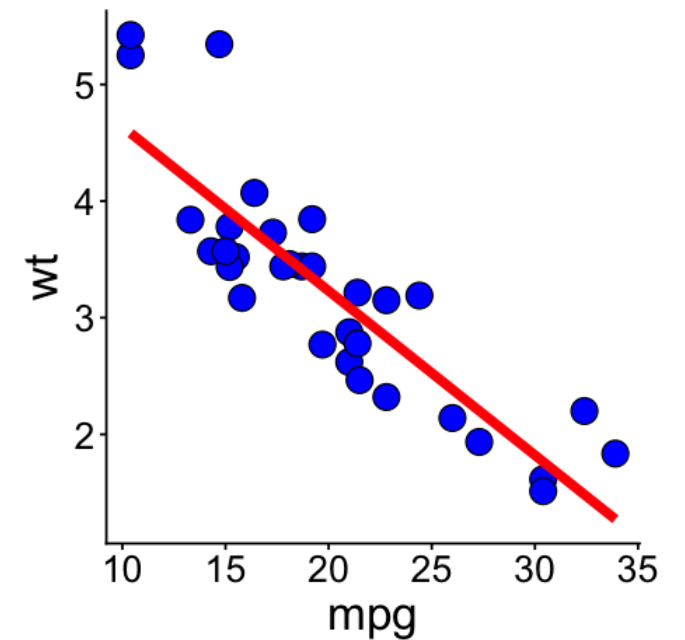


CODE

```
mtcars %>%  
  dplyr::select(mpg, wt) %>%  
  ggplot(aes(x = mpg, wt))+  
  geom_point(fill = "blue",  
             size = 5,  
             shape = 21)+  
  geom_smooth(method = "lm",  
             se = FALSE,  
             color = "red",  
             alpha = 0.2,  
             size = 2)+  
  theme_classic()+  
  theme(text = element_text(size = 20),  
        aspect.ratio = 1)
```



PLOT



Why R?

IT'S FREE!



Annual Academic

\$260
per year

Ideal for individual academics

Buy

IBM SPSS

Base subscription

Starting at

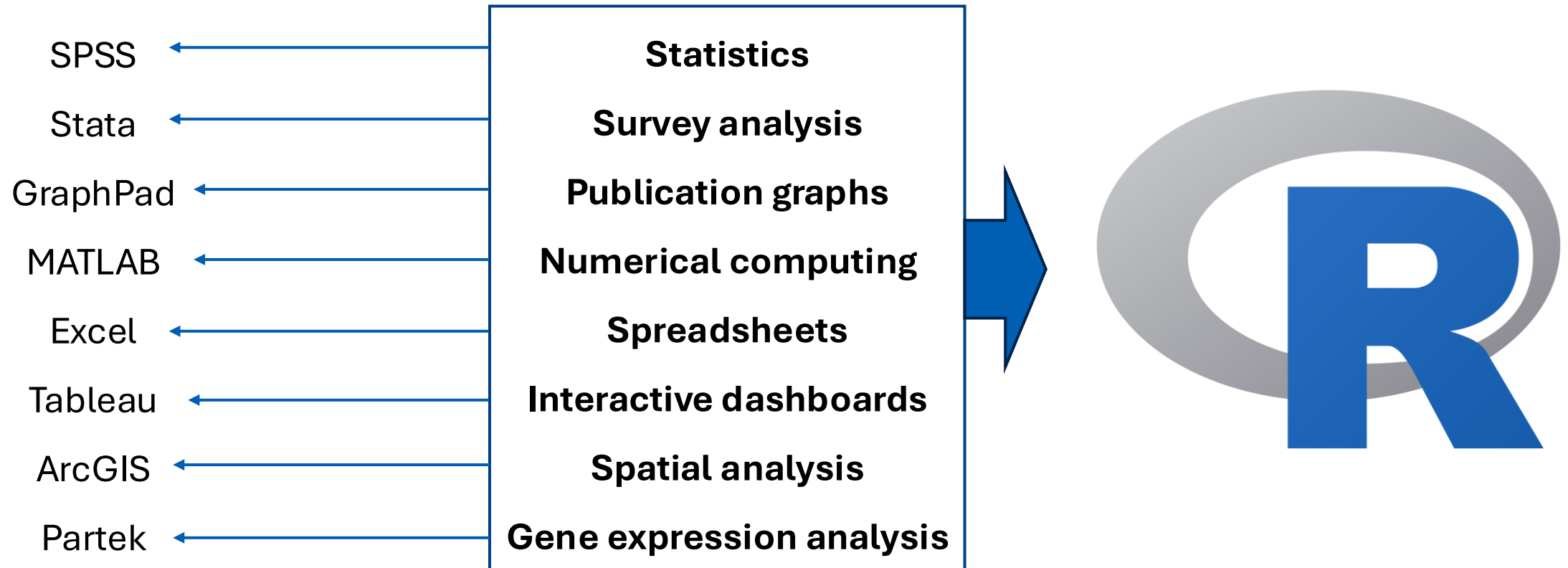
\$109.00

USD Per Authorized User*

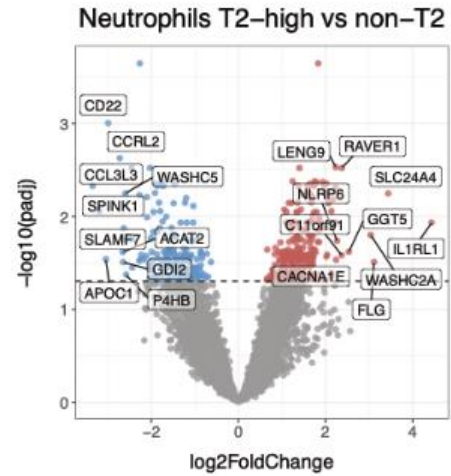
Buy now →

Why R?

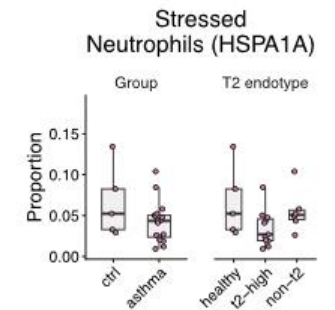
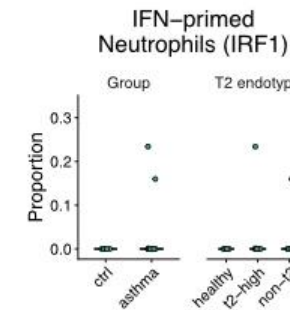
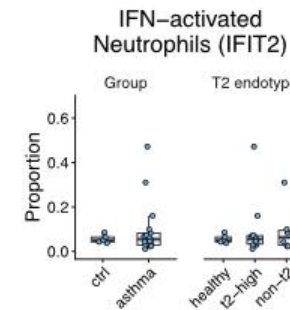
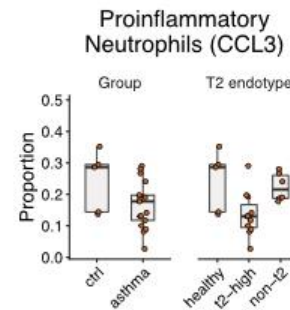
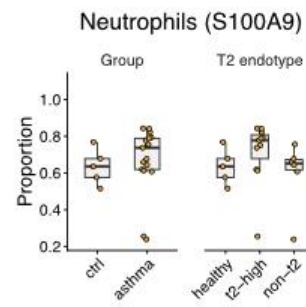
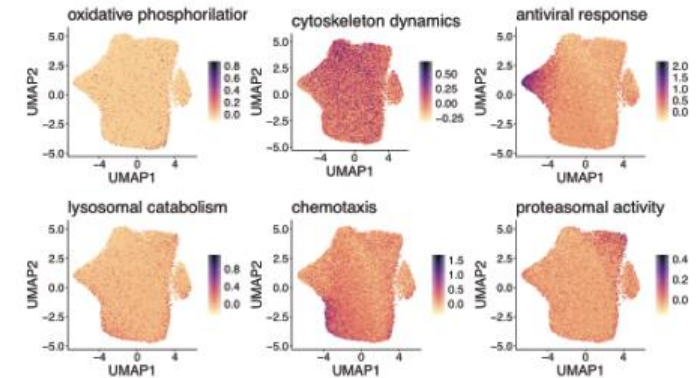
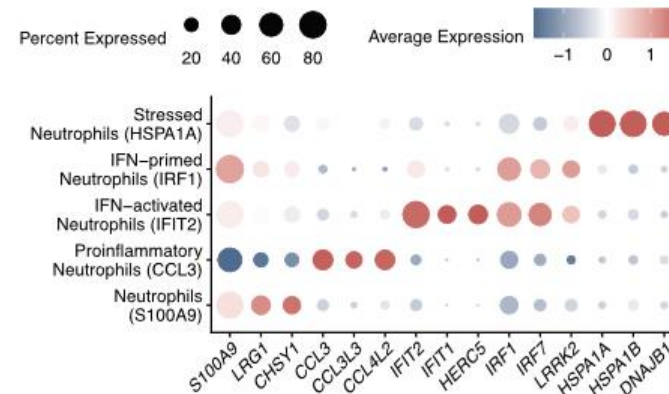
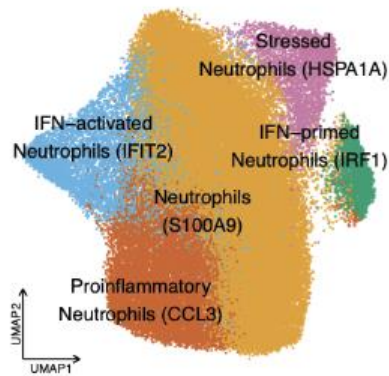
IT'S POWERFUL



Why R?



GSEA GO:BP network



Why R?

BIG, HELPFUL COMMUNITY

R User Community | Meetup.com

890



R User Groups

766364



Members

86



Countries

416

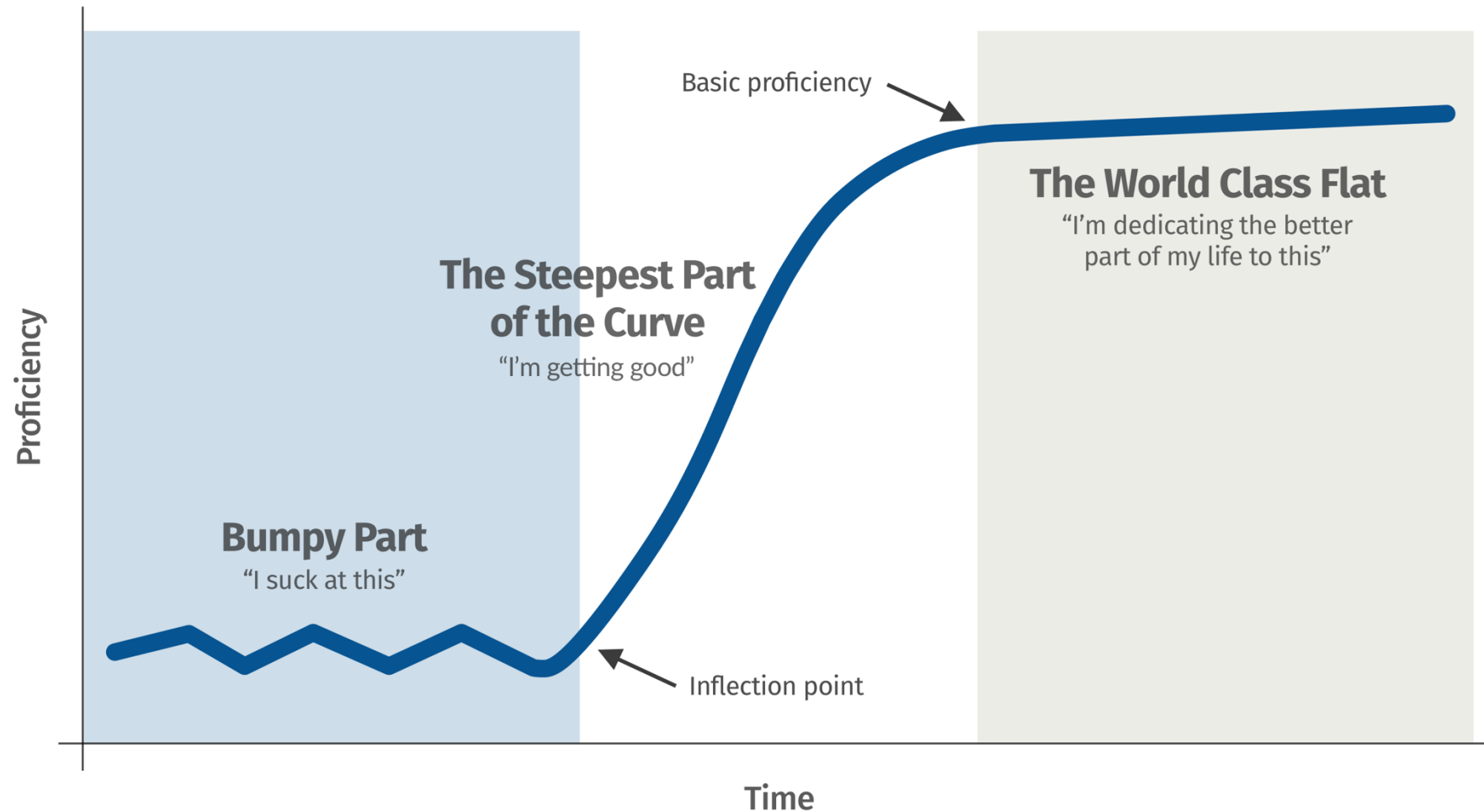


Cities



The main con:

STEEP LEARNING CURVE



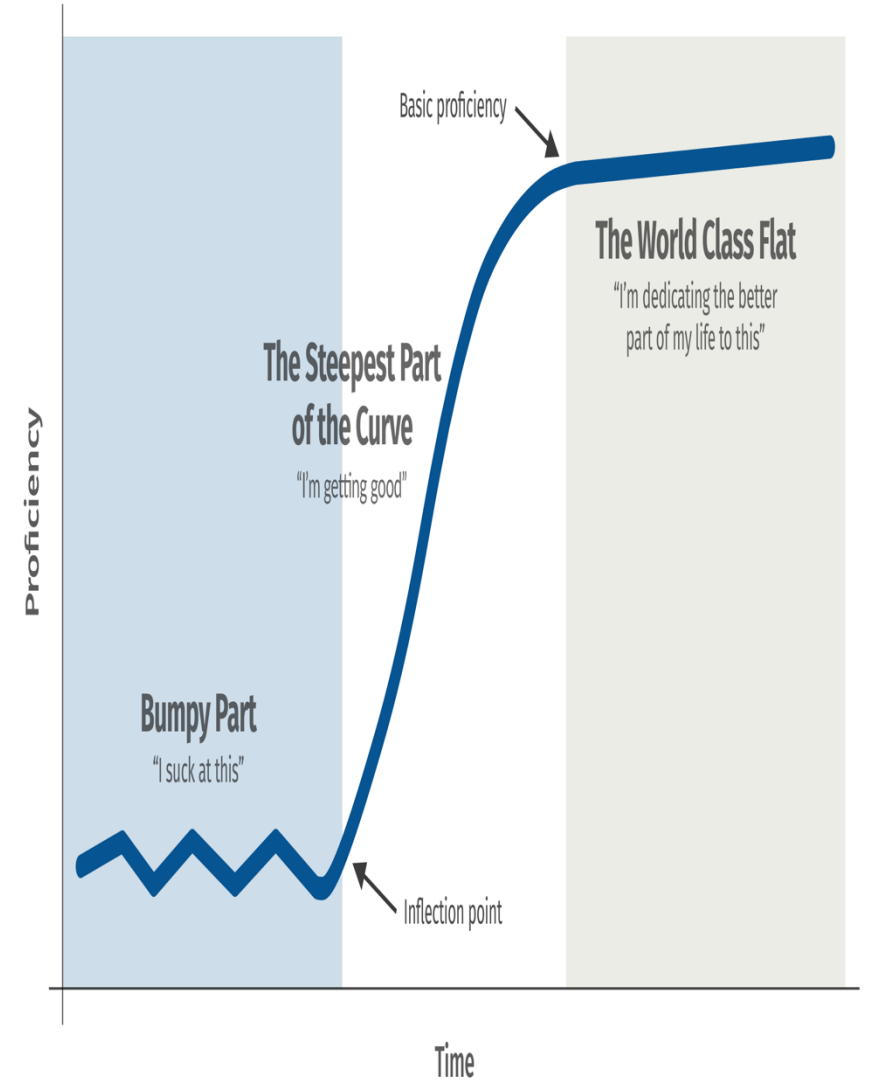
IA



Claude



ChatGPT



Hi Claude, can you give me an R script to produce a dot plot with ggplot2? My dataset is called starwars and I want to plot in the x-axis the variable mass and in the y-axis the variable height. I want the dots to be colored by sex. I'd also like to add a background

We need to know:

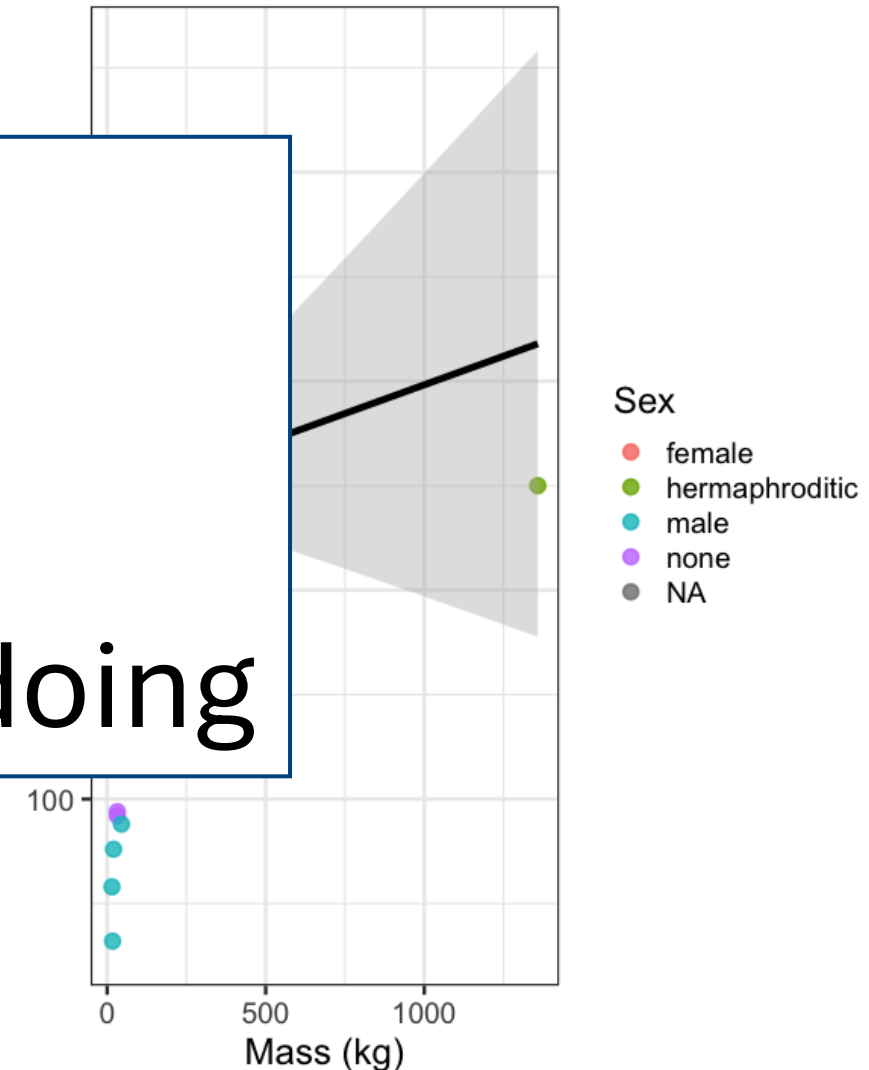
1. What to ask for
2. What the code is doing

Created a file, shared a file, r

Here's the script:

```
r
library(tidyverse)

starwars |>
  filter(!is.na(mass))
  ggplot(aes(x = mass, y = height)) +
  geom_point(aes(color = sex), size = 3, alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, color = "black", fill = "grey70") +
  labs(x = "Mass (kg)", y = "Height (cm)", color = "Sex") +
  theme_bw(base_size = 18)
```



Intro to R GitHub



cecilomar6 / RWIP_intro_to_R ▾

 **RWIP_intro_to_R** Public



 **main** ▾  **1 Branch**  **0 Tags**

 Go to file

T

Add file ▾

<> Code ▾

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358f90e · 1 minute ago  **4 Commits**

 tidyverse_cheat_sheets

Add files via upload

34 minutes ago

 README.md

Update README.md

1 minute ago

 **README**



RWIP_intro_to_R

Materials for Intro to R lessons

Step 1: Install R

<https://cran.rstudio.com/>

Step 2: Install RStudio

<https://posit.co/downloads>

Step 1: Install R

<https://cran.rstudio.com/>

The Comprehensive R Archive Network

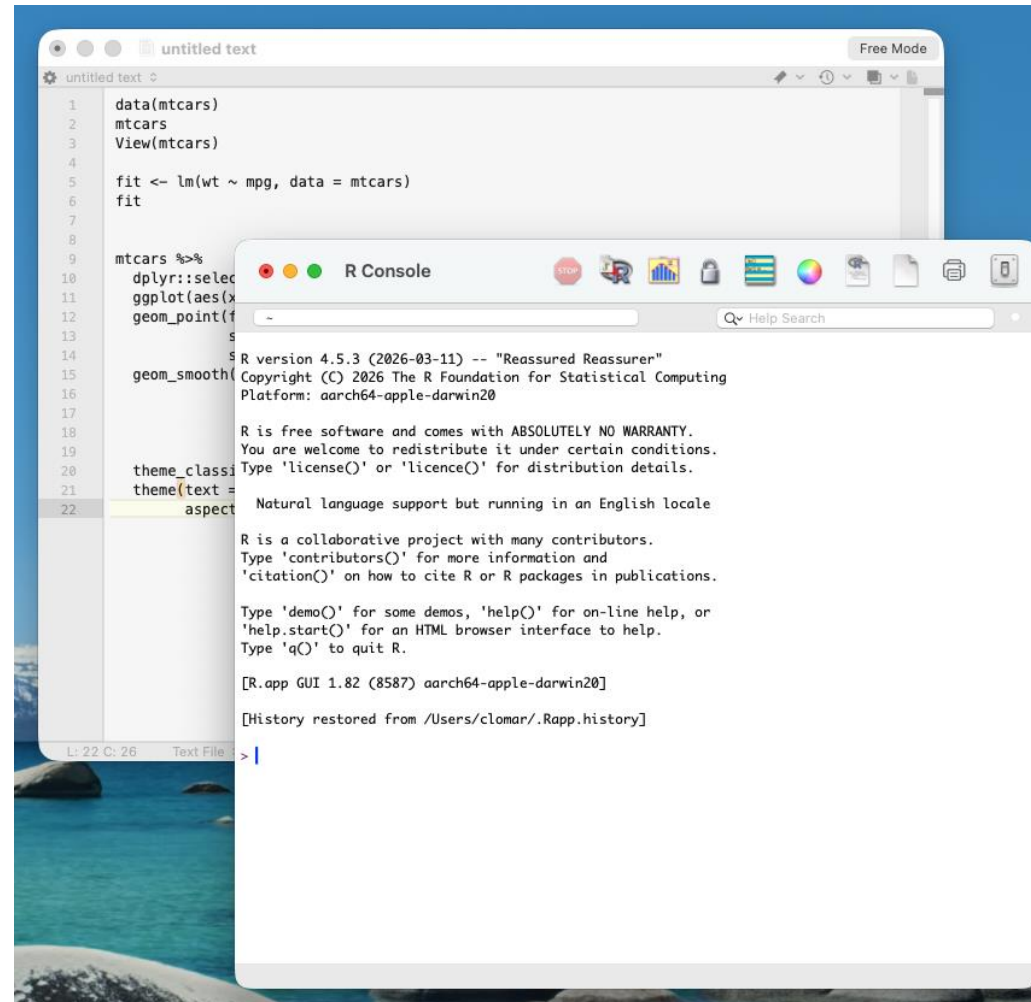
Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#) ([Debian](#), [Fedora/Redhat](#), [Ubuntu](#))
- [Download R for macOS](#)
- [Download R for Windows](#)

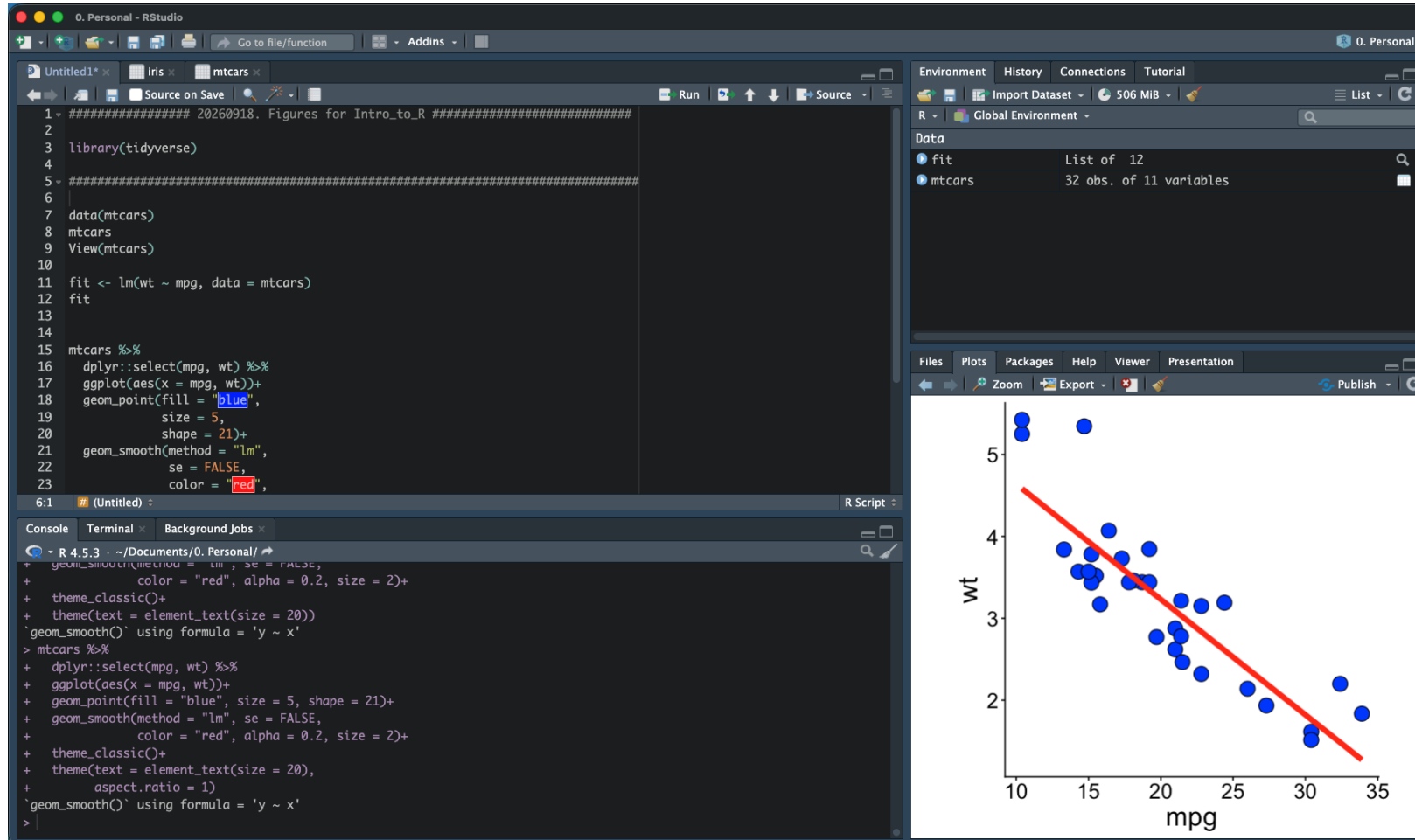
R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

What R might look like



What is Rstudio?

An interface for interacting with and executing R commands



Step 2: install RStudio

<https://posit.co/downloads>

DOWNLOAD

RStudio IDE

The most popular coding environment for R,
built with love by Posit.

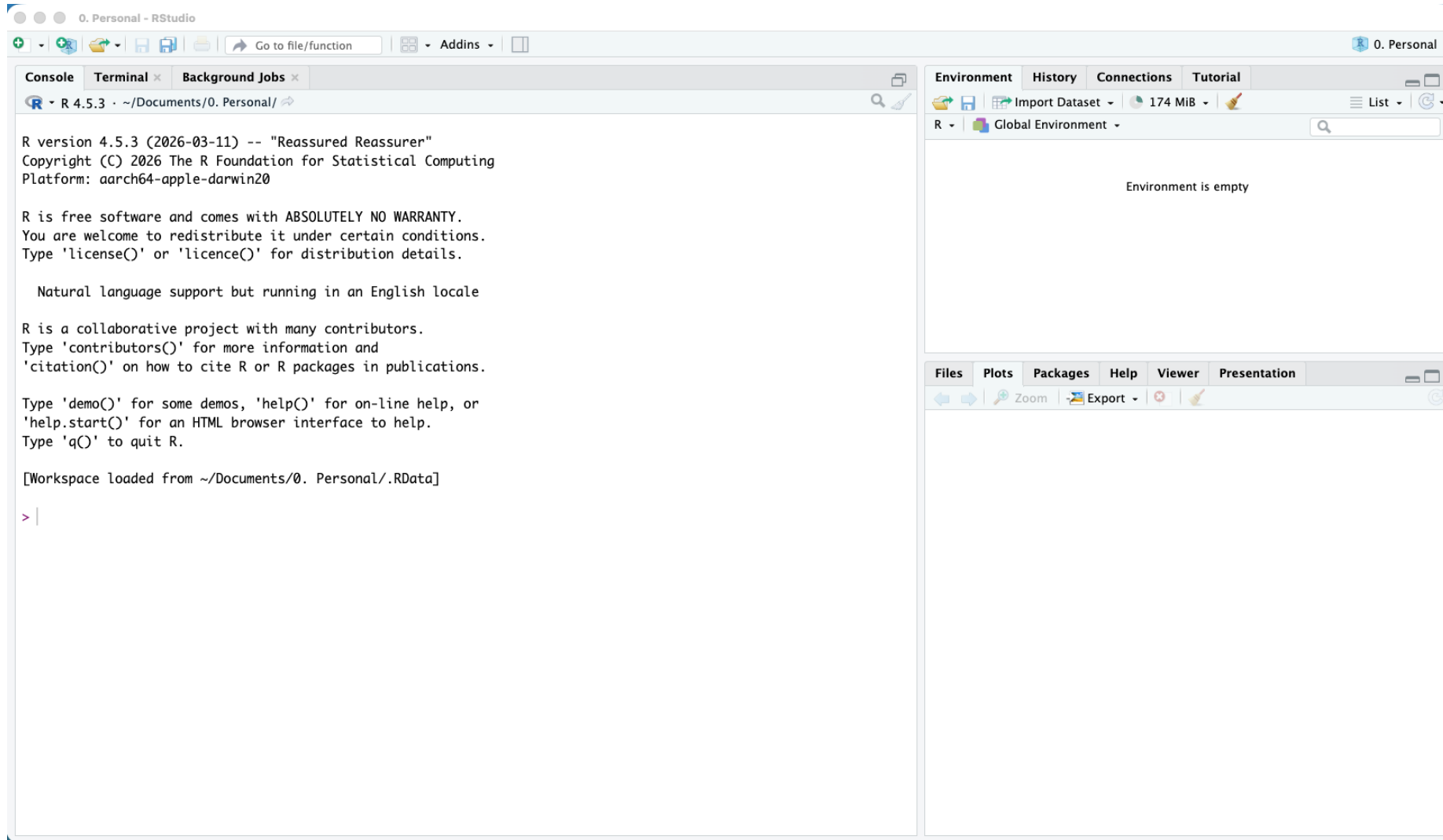
Used by millions of people weekly, the RStudio integrated development environment (IDE) is a set of tools built to help you be more productive with R and Python. It includes a console, syntax-highlighting editor that supports direct code execution. It also features tools for plotting, viewing history, debugging and managing your workspace.

If you're a professional data scientist and want guidance on adopting open-source tools at your organization, don't hesitate to book a call with us.

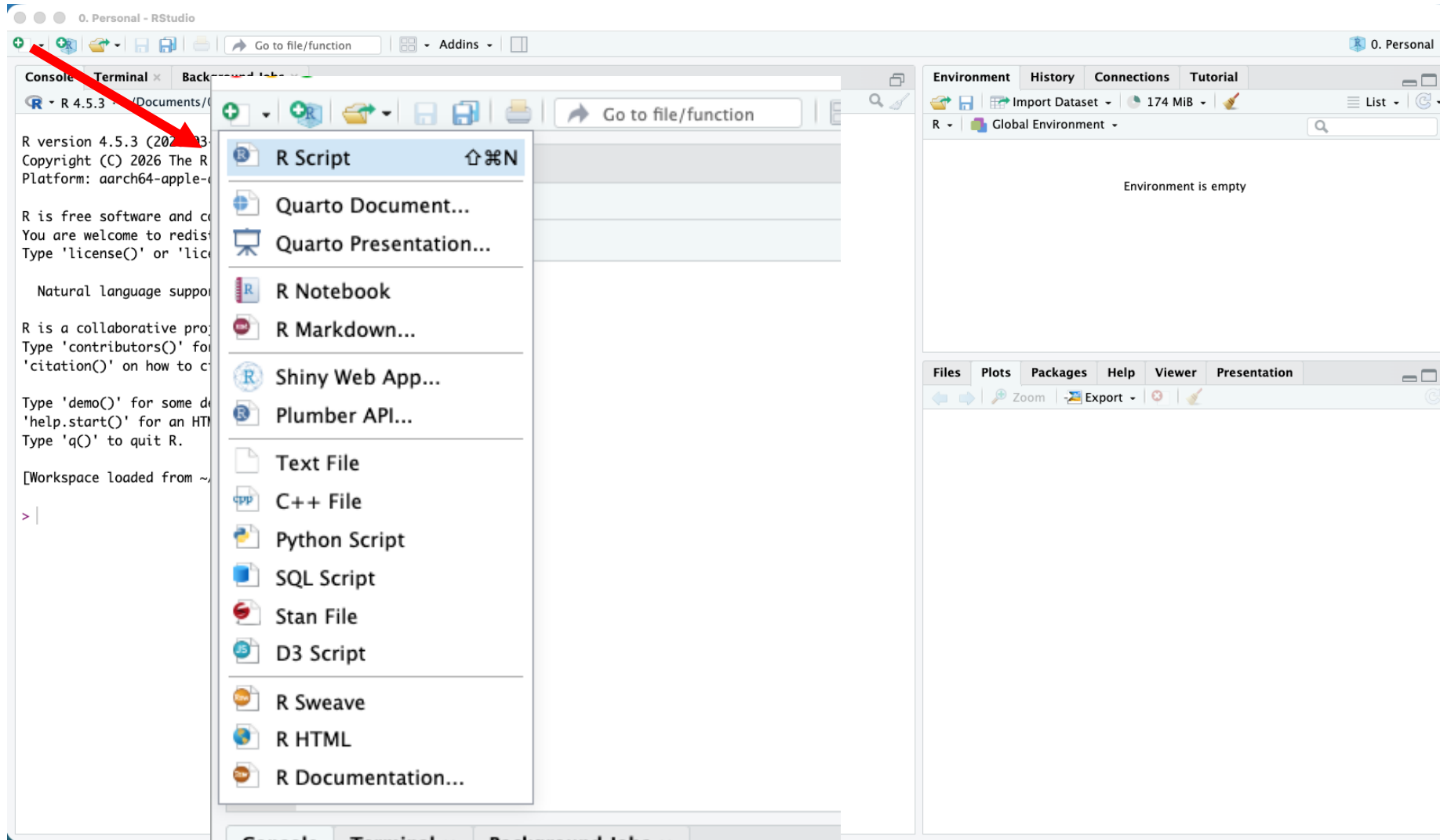
DOWNLOAD RSTUDIO

DOWNLOAD RSTUDIO SERVER

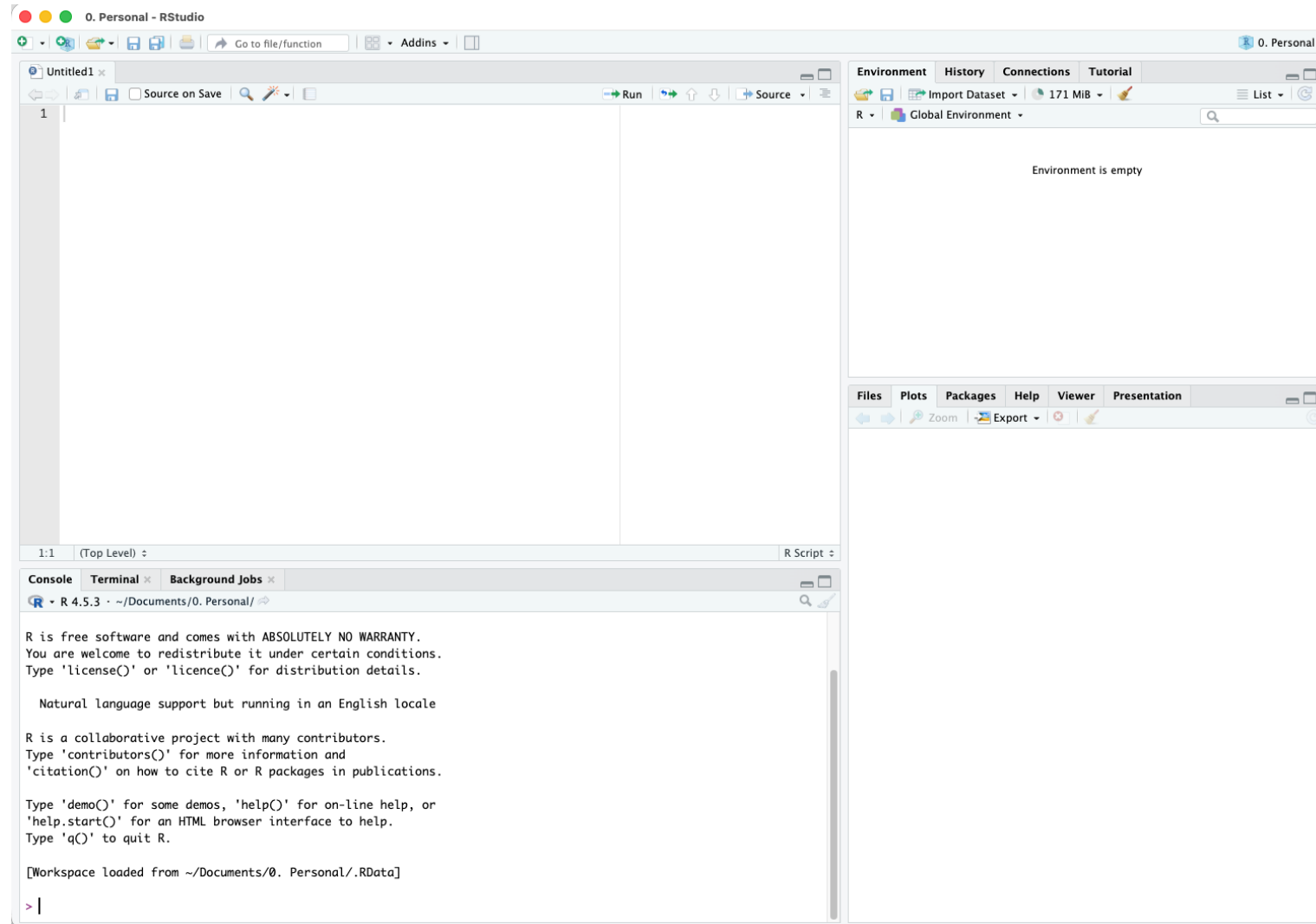
Installing R and Rstudio



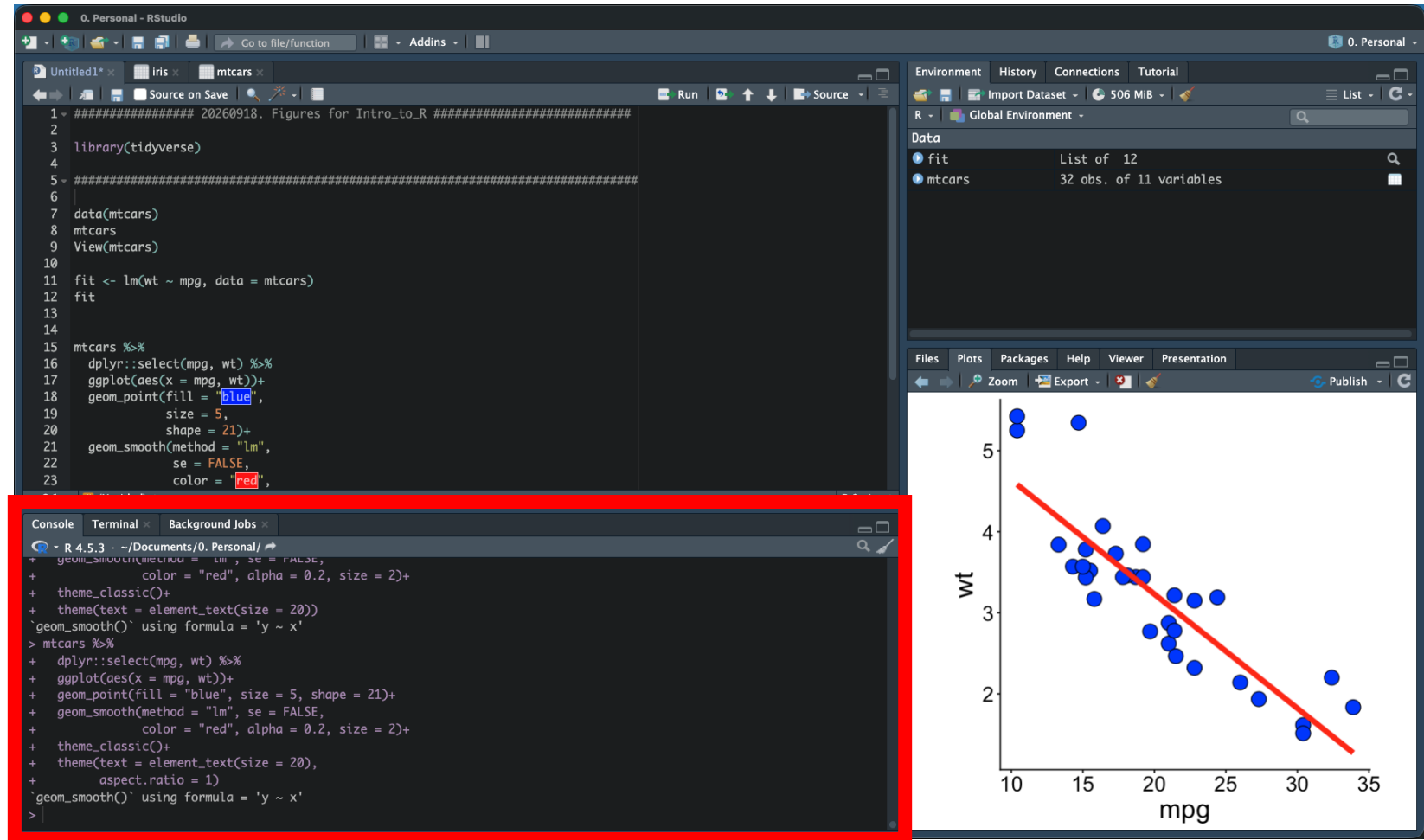
Installing R and Rstudio



Installing R and Rstudio

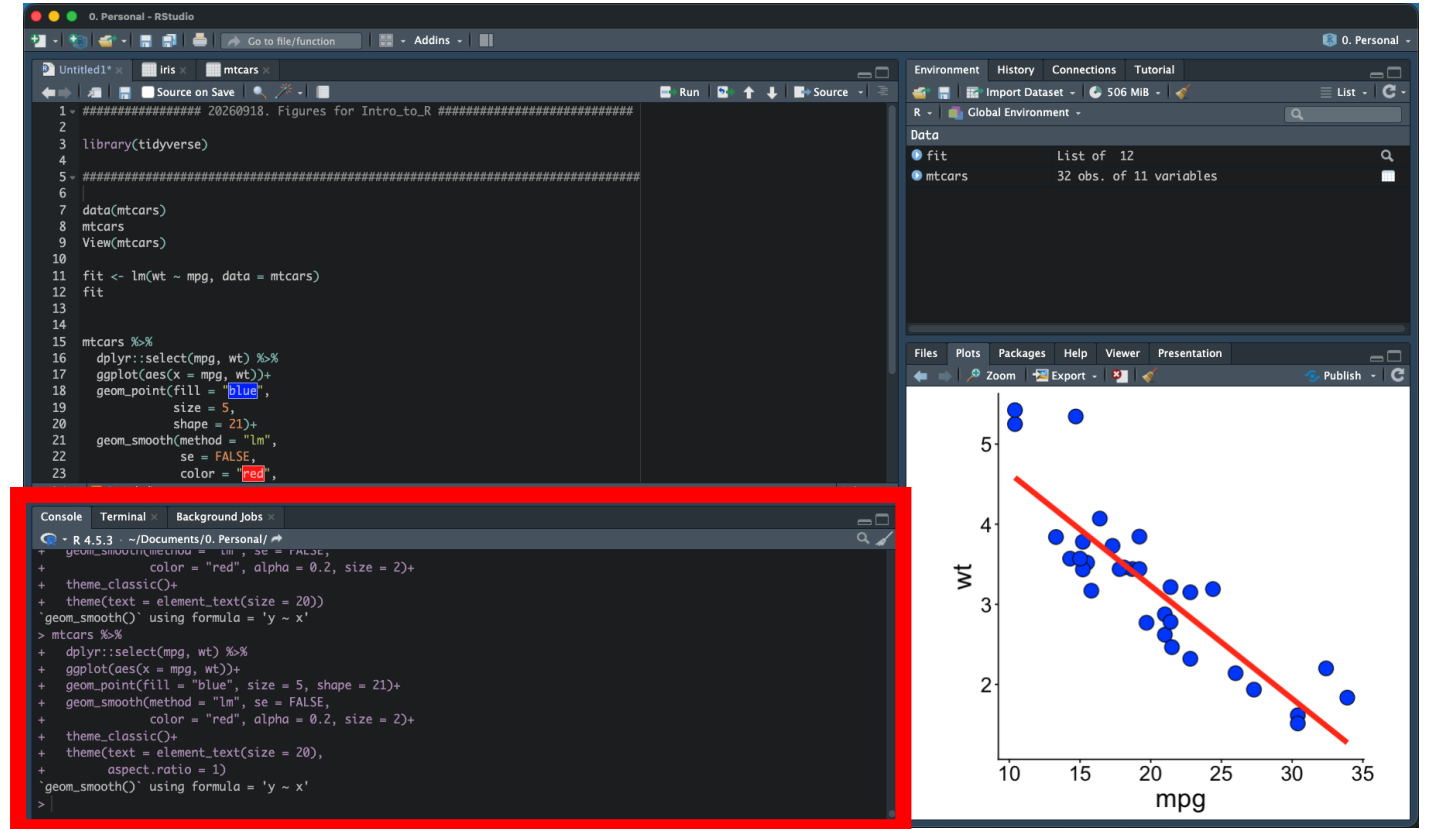
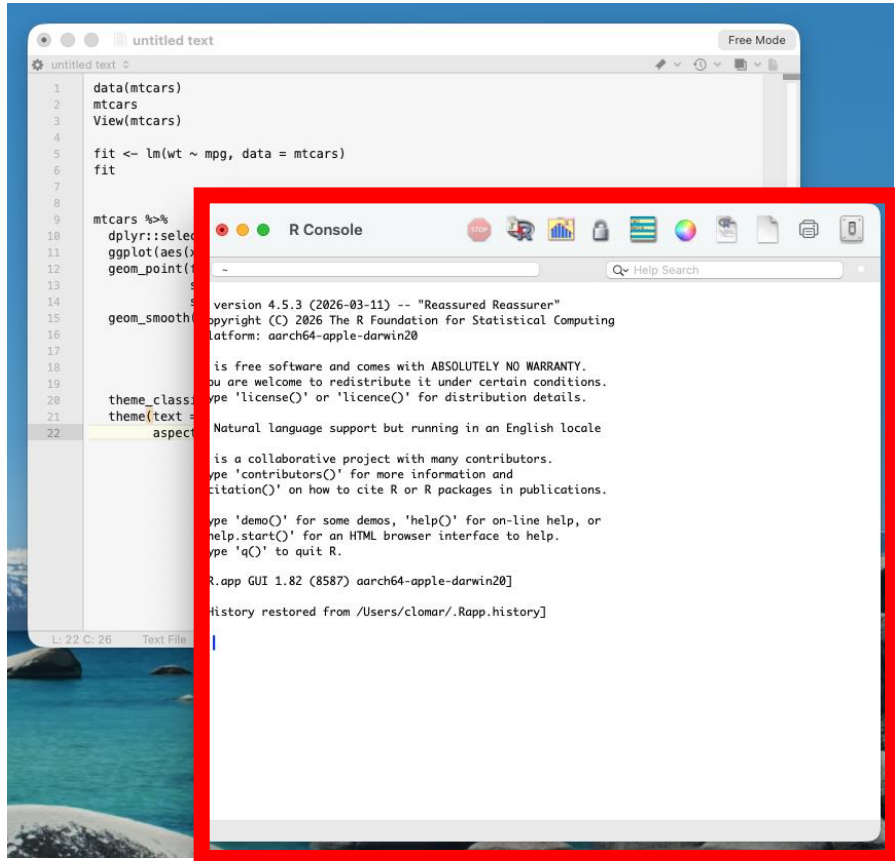


Rstudio overview



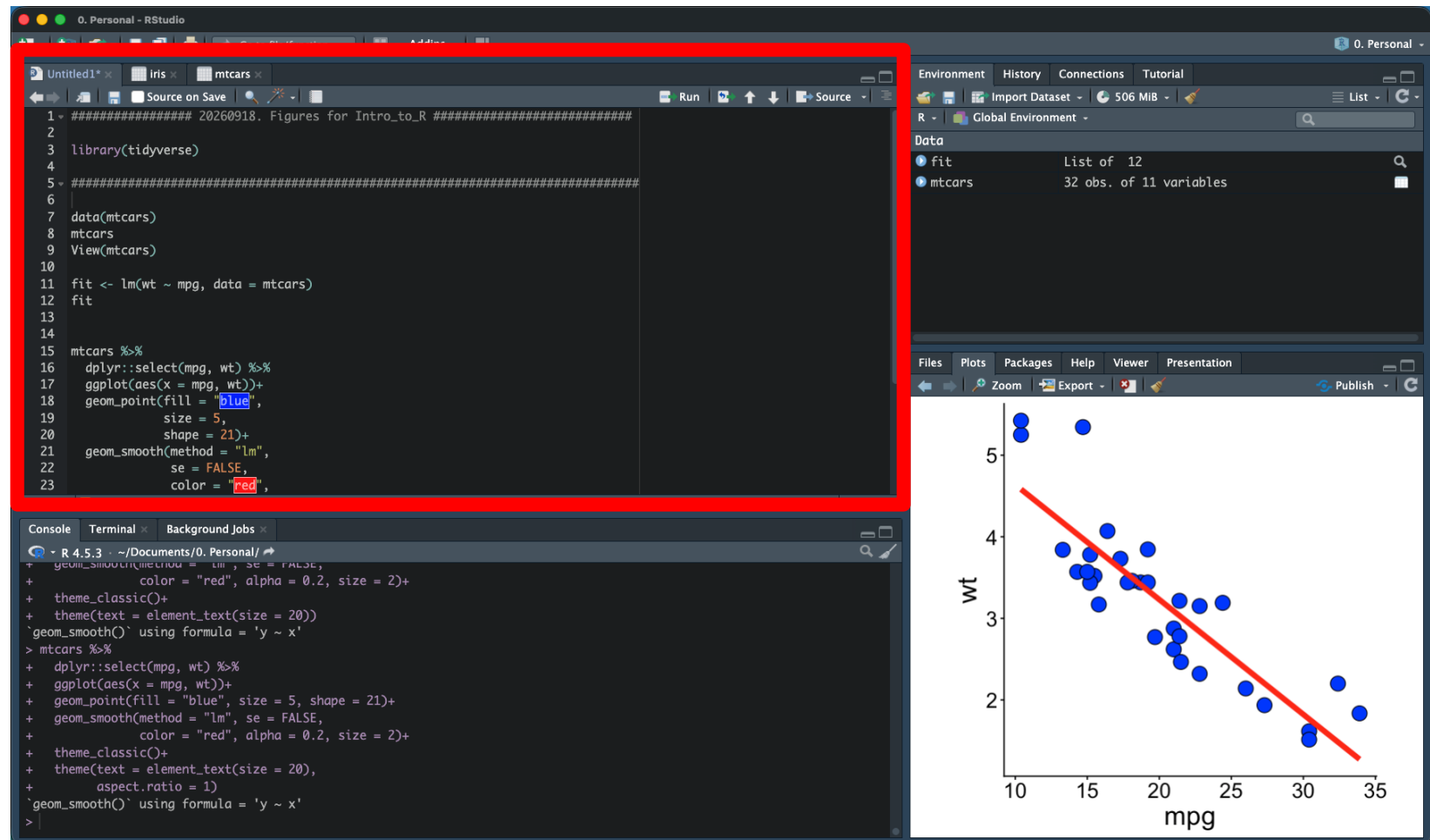
Console Pane:
R terminal.
Shows what R
commands have
been run

Rstudio overview

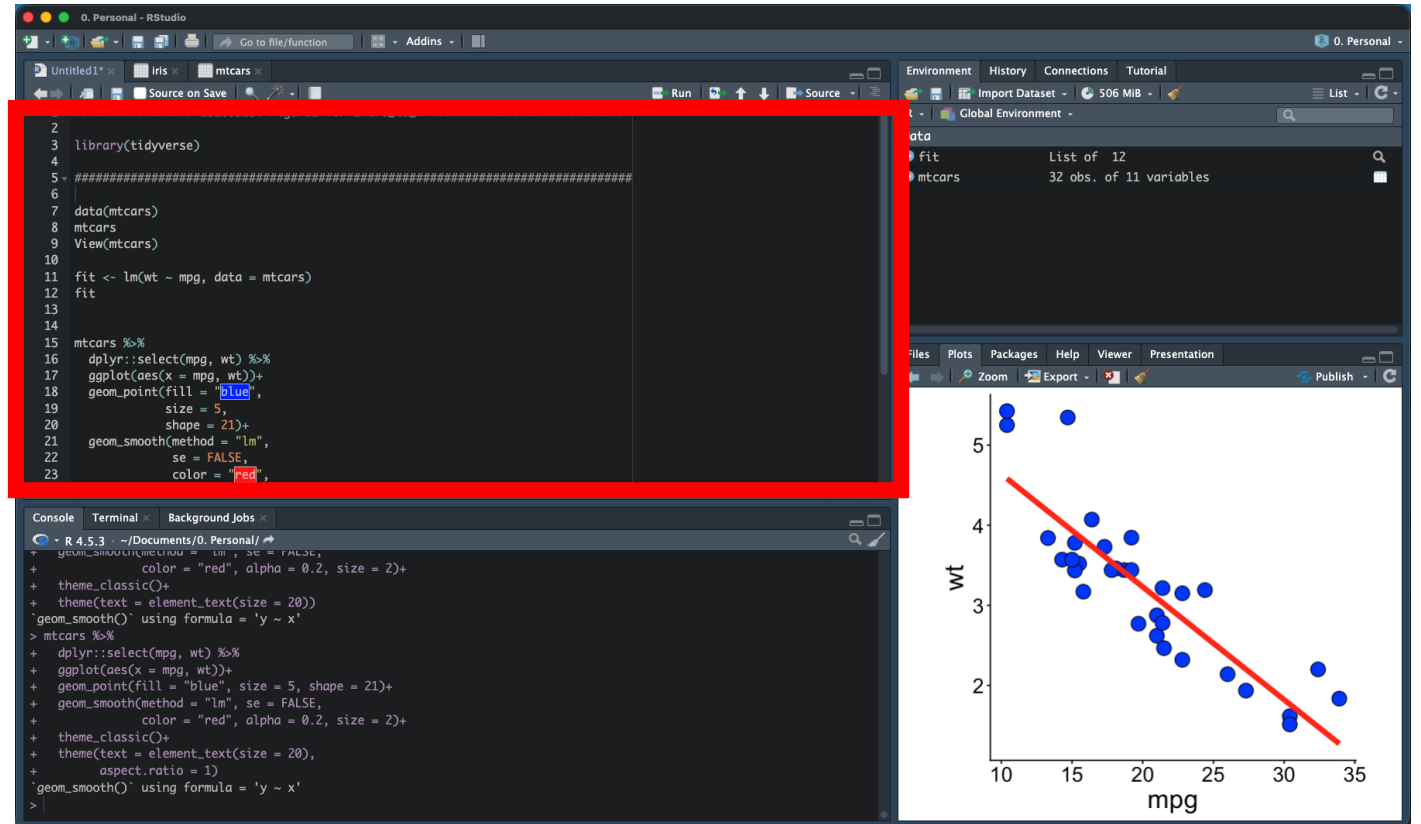
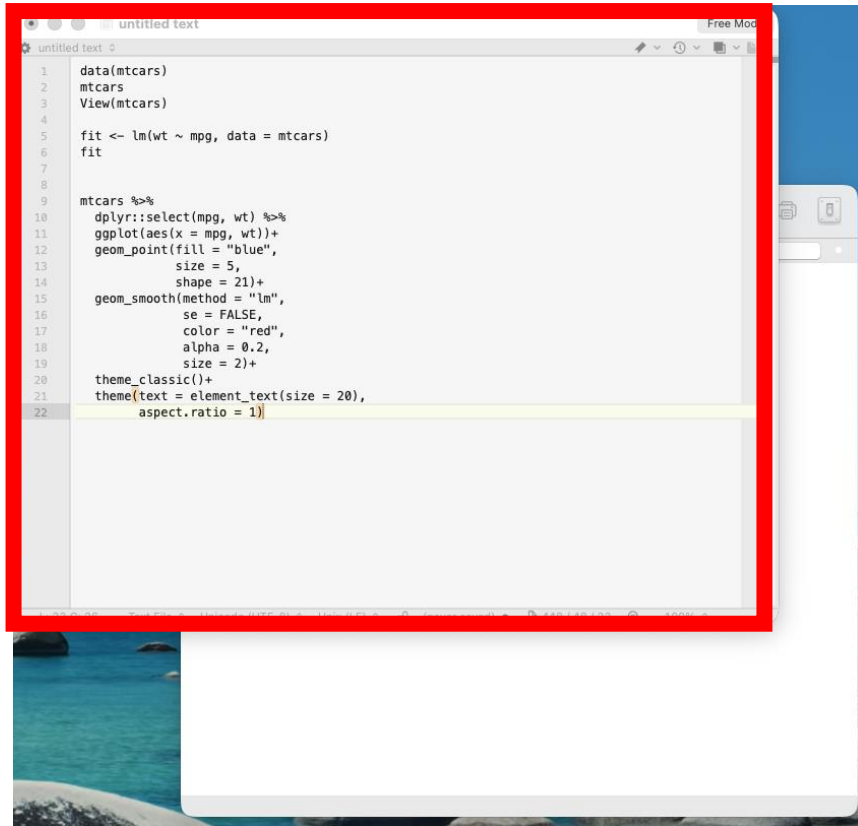


Rstudio overview

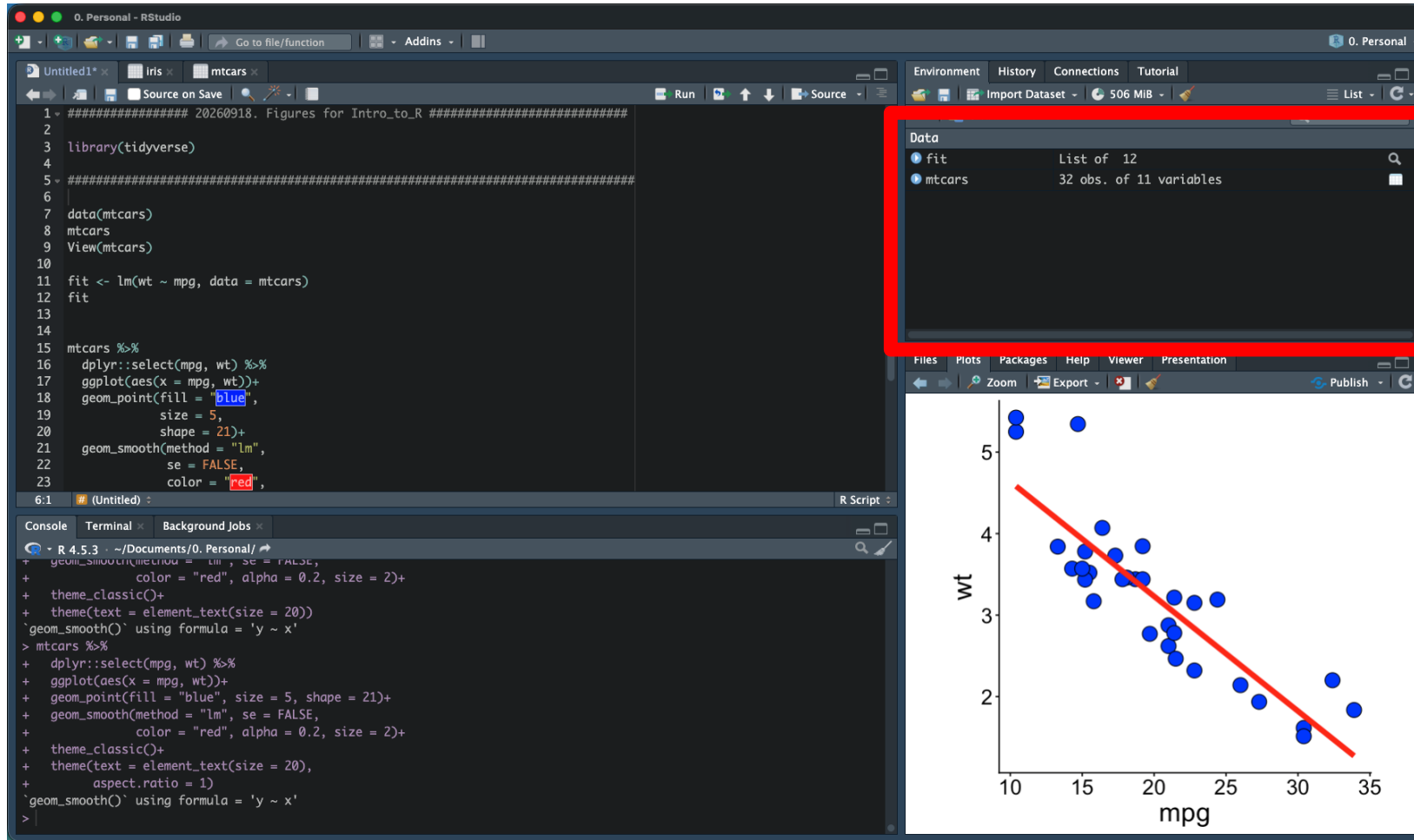
Source Pane:
Files and data
tables



Rstudio overview

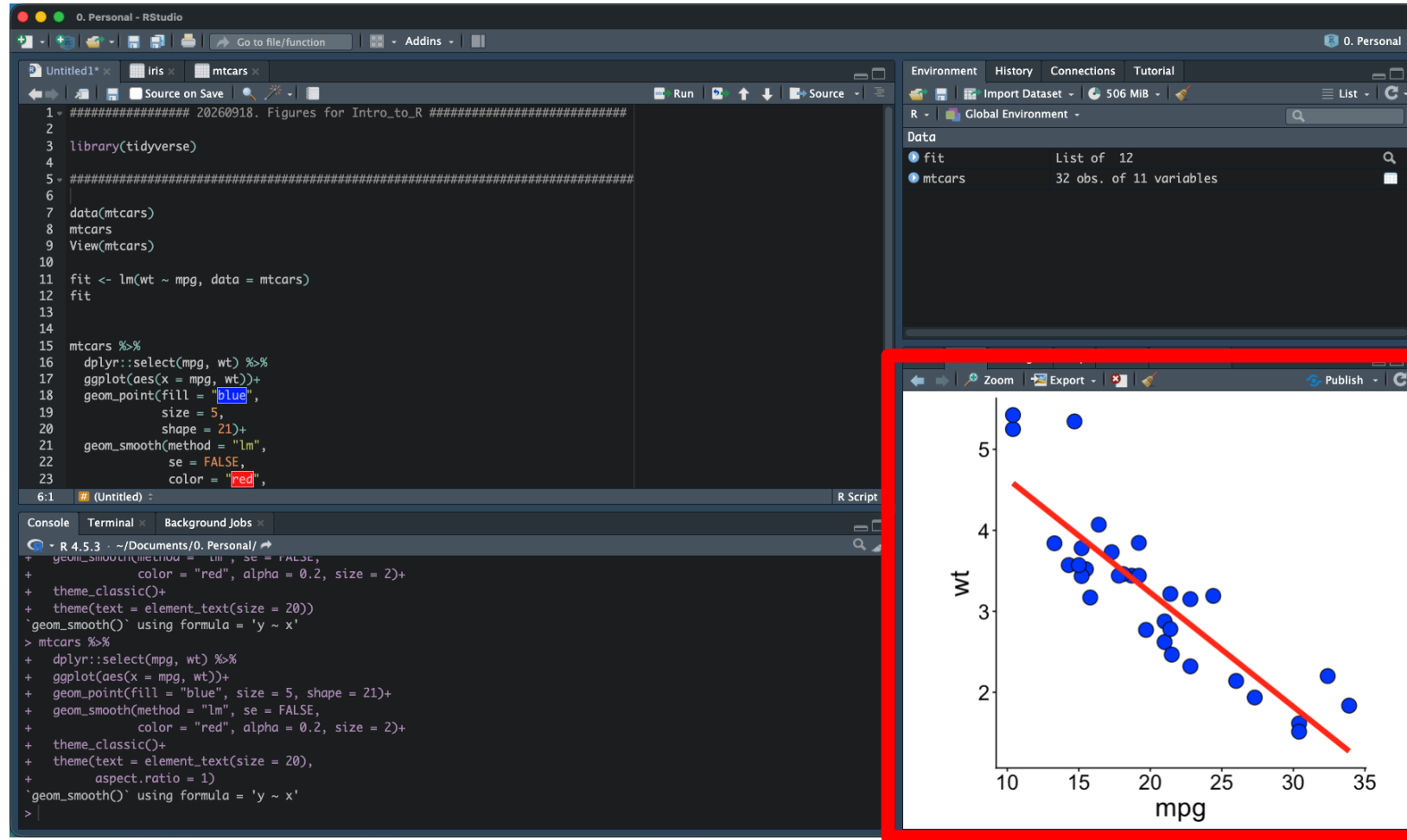


Rstudio overview



**Environment
Pane:** Data
tables that are
currently in your
environment.

Rstudio overview



**Files & Plots
Pane:** Shows
files in a
directory

The R language

English:

Subject Verb Object

R:

Function(Object)

The R language

English:

Subject Verb Object

R:

Function(Object)

The R language

English:

Subject **Verb** **Object**

“Ceci filters the dataset”

R:

Function(**Object**)

filter(dataset)

The R language

English:

Subject Verb Object

Adjectives

Adverbs

Prepositions

R:

Function(Object, Arguments)

Objects

- Named container holding some data
- Its value is assigned by the symbol “<-”

```
> x <- 45  
> x  
[1] 45  
>
```

```
> y <- "apple"  
> y  
[1] "apple"  
>
```

Objects

0-dimensional

```
> x <- 45
```

```
> y <- "apple"
```

1-dimensional

```
> x <- c(1,2,3)
```

```
> x
```

```
[1] 1 2 3
```

```
> y <- c("apple", "pear", "banana")
```

```
> y
```

```
[1] "apple" "pear" "banana"
```

VECTORS

2-dimensional

```
> z <- data.frame(x,y)
```

```
> z
```

```
  x      y
1 1  apple
2 2   pear
3 3 banana
```

DATA FRAMES or **MATRICES**

Shortcut to write "<-"
Option + -

Objects

Mixed

S4 OBJECT

LISTS

FITTED MODEL

FACTOR

Objects

Objects have a specific **class**

```
> x <- 45  
> class(x)  
[1] "numeric"
```

```
> y <- "apple"  
> class(y)  
[1] "character"
```

```
> x <- "45"  
> class(x)  
[1] "character"
```

Fundamental classes: numeric, integer, character, logical

Objects

Specific functions require objects of a specific class

```
> sum(45, 46)  
[1] 91
```

```
> sum("45", "46")  
Error in sum("45", "46") : invalid  
'type' (character) of argument
```

Functions

Named block of code that:

1. Takes an input
2. Does something with those inputs
3. Returns an output

Functions

```
> sum(45, 46)  
[1] 91
```

1. Takes an
input

(45, 46)

2. Does something
with those inputs

45 + 46

3. Returns an
output

[1] 91

Functions

We can assign the output of a function to an object:

```
> x <- sum(45, 46)  
> x  
[1] 91
```

Functions

Some functions have a specific character assigned to them, **OPERATORS**

```
> sum(45, 46)
[1] 91
```

```
> 45 + 46
[1] 91
```

+	addition
-	subtraction
*	multiplication
/	division
^	exponent
%%	modulo (remainder)
%/%	integer division

<-	assign (standard)
[]	subset, keep structure
[[]]	extract single element
\$	access by name (lists, data frames)
@	access a slot (S4 objects specifically)
:	sequence generator, e.g. 1:5
~	formula operator, e.g. y ~ x

Packages

Bundle of functions that can be downloaded from a repository:

- **CRAN:** `install.packages("....")`
- **Bioconductor:** `BiocManager::install(".....")`
- **GitHub:** `devtools::install_github("user/repo")`

To use them they need to be activated:

```
> library("dplyr")
```



Tidyverse

R packages for data science

The tidyverse is an opinionated **collection of R packages** designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

Install the complete tidyverse with:

```
install.packages("tidyverse")
```

Intro to R GitHub



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 RWIP_intro_to_R Public

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c3ef2cb · 1 minute ago 5 Commits

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 20260930_intro_to_r.R	Add files via upload	1 minute ago
 README.md	Update README.md	last week

README

RWIP_intro_to_R

Materials for Intro to R lessons

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About

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cecilomar6

Packages



Learn R, in R.

swirl teaches you R programming and data science
interactively, at your own pace, and right in the R
console!

Swirl

```
> install.packages("swirl")
```

```
> library("swirl")
```

```
> swirl()
```

```
| Welcome to swirl! Please sign in. If you've been  
| here before, use the same name as you did then.  
| If you are new, call yourself something unique.
```

```
What shall I call you? Ceci|
```

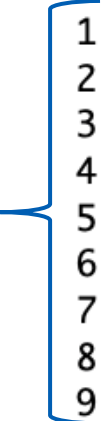
Swirl

| Please choose a course, or type 0 to exit swirl.

- 1: R Programming
- 2: Take me to the swirl course repository!

Selection: 1|

| Please choose a lesson, or type 0 to return to
| course menu.

- 
- 1: Basic Building Blocks
 - 2: Workspace and Files
 - 3: Sequences of Numbers
 - 4: Vectors
 - 5: Missing Values
 - 6: Subsetting Vectors
 - 7: Matrices and Data Frames
 - 8: Logic
 - 9: Functions
 - 10: lapply and sapply
 - 11: vapply and tapply
 - 12: Looking at Data
 - 13: Simulation
 - 14: Dates and Times
 - 15: Base Graphics

Selection: 1